

Lesson 4: Create PT Written Response Mock Exam

90 minutes

Overview

How can I employ effective strategies to excel in the AP Computer Science Principles written exam?

On Day 1 of the lesson, students explore the exam format, learn test-taking and annotation strategies, and finally craft responses to two sample Performance Task prompts, specifically addressing #1 "Program Design, Function, and Purpose" and #2(a) "Algorithm Development." Students should be prepared with completed Personalized Project Reference (PPR) sheets. The lesson emphasizes discussions, annotations, and writing exercises to build confidence and strategic skills for the upcoming exam.

On Day 2 of this lesson, the teacher models the day 1 strategies by annotating a sample performance task question. Then, students begin the second half of the mock exam while the teacher facilitates, encourages, and helps clarify any confusion. Finally, students participate in quick partner and whole class review to reinforce collaboration, acknowledge challenges, and make suggestions for future sessions.

Standards

Full Course Alignment

CSP Conceptual Framework

- **AAP-1** - To find specific solutions to generalizable problems, programmers represent and organize data in multiple ways.
- **AAP-2** - The way statements are sequenced and combined in a program determines the computed result. Programs incorporate iteration and selection constructs to represent repetition and make decisions to handle varied input values.
- **AAP-3** - Programmers break down problems into smaller and more manageable pieces. By creating procedures and leveraging parameters, programmers generalize processes that can be reused. Procedures allow programmers to draw upon existing code that has already been tested, allowing programmers to write programs more quickly and with more confidence.

Objectives

Students will be able to:

- Craft a response to four sample Create PT Prompts
- Identify and incorporate a variety of strategies for answering free response questions.

Preparation

- Print Personalized Project References
- Choose one, or both
 - Print copies of Activity Guide (Version A) - Create PT Mock Exam (one for each student)
 - Print copies of Activity Guide (Version B) - Create PT Mock Exam (one for each student)
- Print and cut Create PT Mock Exam Description
- Set up a space for the "Want to Know" area that students can contribute to
- Gather sticky notes for each student to use
- Check the "[Teacher's Lounge](#)" forum for verified teachers to find additional strategies or resources shared by fellow teachers

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [AP CS Principles Course and Exam Description \(CED\)](#) - Resource

- ▶ **AAP-4** - There exist problems that the computer cannot solve, and even when the computer can solve a problem, it may not be able to do so in a reasonable amount of time.
- ▶ **CRD-2** - Developers create and innovate using an iterative design process that is user-focused, that incorporates implementation/feedback cycles, and that leaves ample room for experimentation and risk-taking.

Agenda

Day One (45 minutes)

Warm Up

Activity

Wrap Up

Day Two (45 minutes)

Warm Up

Activity

Wrap Up

- **AP Classroom** - Resource
- **Unit 9 Lesson 4 - Create PT Mock Exam - Slides** - Slides

▼ Make a Copy

For the students

- **Activity Guide (Version A) - Create PT Mock Exam** - Handout
▼ Make a Copy
- **Activity Guide (Version B) - Create PT Mock Exam** - Handout
▼ Make a Copy
- **Create PT Mock Exam Description** - Handout
▼ Make a Copy

Teaching Guide

Day One (45 minutes)

Warm Up (5 minutes)

Review Exam Requirements

💡 Teaching Tip ▲

The prompts in this lesson come from the CED and are publicly available. Some teachers may wish to use alternate prompts. There are three practice exams available only to teachers in the “Resources and Assignments” area of **AP Classroom**. These practice exams provide comprehensive coverage of the written response portion of the Create Performance Task, as well as scoring guidelines for each of the four prompts. If a teacher chooses to use one or more of these prompts, they should replace the prompts in the Activity Guide. Prompts appear on pages 2-6 of Version A and pages 2 and 4-7 of Version B.

🎤 Remarks

Today you will explore strategies for the free response portion of the AP CS Principles exam. By the end of the next class, you will have answered all four parts required for the free response portion of this mock exam.

Distribute: Create PT Mock Exam Description

Display: Instructions on the Quick Annotations Slide.

Do This: Ask students to circle key terms and underline anything unclear. Encourage brief notes or questions in the margins.

Distribute: Sticky notes for Parking Lot

 **Display:** Instructions on the Parking Lot slide

Do This: Ask students to jot down any questions or points of confusion, and place them in the "Want to Know" area.

 Teaching Tip

"**Want to Know**" comes from K-W-L charts, wherein students describe what they already know about a topic, what they want to know about a topic, and then after reading, what they have learned. Some classes may be more comfortable with this format.

After students place items in the "Want to Know" area take a minute to group similar things together to help you identify themes. One simple way to do this if nothing else becomes obvious is group them into: Ideas, Concerns, Questions. This will help you refer back to items in this area as themes and questions get addressed. Then quickly move into the activity of the lesson.

Activity (35 minutes)

Review Strategies

 **Discuss:** *What strategies have you used before to help you plan and write your answers to free-response questions?*

Discussion Goal: Use this discussion to allow students to share any strategies they have used in the past, even from other subjects. You may also push students to think about things they wish they had done in approaching the practice they've encountered throughout the year.

Distribute: *Personalized Project References (PPR)*

Choose one of the activity guide options:

- **Activity Guide (Version A) - Create PT Mock Exam**
- **Activity Guide (Version B) - Create PT Mock Exam**

 Teaching Tip

Differentiation Opportunity

In recognizing the diverse learning preferences and needs of our students, we present two annotation strategies: Version A and Version B. **Version A** involves circling key terms in the question for a focused response. It's a concise approach suitable for those who prefer simplicity and efficiency.

Version B, on the other hand, adds a more detailed layer to the process. It includes circling key terms, numbering them, writing short summaries, and freewriting ideas. This method aims to stimulate creativity and build a comprehensive resource during response writing.

Teachers and students are encouraged to choose the annotation strategy that aligns best with their preferences, learning styles, and objectives.

Remarks

Answering free response questions on test day can be intimidating, but it doesn't have to be. Today you will learn a variety of strategies for planning, organizing, and writing each of your responses.

Do This: Turn to page 1 of the activity guide and review the strategies. Have students read through the strategies, and ask clarifying questions as needed.

Teaching Tip

While it is important to expose students to these strategies, don't spend too long explaining each of them. Practicing with the strategies, and then reflecting on them and asking questions about them, will be a much more effective way of checking for understanding.

Planning

 **Display:** "Planning" section instructions slide.

Remarks

When you get your exam, you will see two questions that make up a total of four prompts. On exam day, you will have 60 minutes to complete all four prompts. Today, you will only answer two questions.

Before answering the questions, it will be important to read each question, and prioritize them based on your confidence level, strengths, and difficulty.

Circulate: The teacher should ensure students complete the planning page in their activity guide rather than jumping straight into answering the questions.

Teaching Tip

For those with a bit more time, this activity can be a powerful way to engage students in the planning part of the process.

The teacher designates areas of the room for each question. Students use sticky notes to jot down their "priority level" (1-4) for each, and posts their notes in the designated area. The teacher has students quickly share their reasoning for prioritizing each question, rotating students through each area and reminding students that it is okay to change their priorities if they hear a convincing argument.

Annotating and Writing

 **Display:** The "Annotating and Writing" instructions slide.

Version A and B of this activity are presented below. Teachers may choose one or both options depending on the needs of the learners.

► **Version A: Streamlined Approach**

► **Version B: Detail-Oriented Approach**

Remarks

Annotate the key terms for your first priority question. Once you've finished the annotations, begin writing. Make sure that, as you write, you are crossing off the key terms you annotated as you address them in your response.

Circulate: Monitor the room as students annotate and write. Encourage students to use the annotation strategies before they begin their response.

Teaching Tip

Be sure to stress to students that they will only be answering two questions. Attempting to answer all four questions in the limited timeframe will likely result in superficial, shallow responses.

Wrap Up (5 minutes)

Retrieve-Pair-Share (2 min)

Remarks

You have made good progress today, and are halfway done with the free response portion of this mock exam. We will conclude tomorrow with the remaining two questions. To end today, let's reflect on what has been done so far.

 **Display:** the "Retrieve-Pair-Share" instructions slide.

Do This: After the practice session, provide time for students to reflect on their annotation and response strategies.

Revisit Parking Lot (3 min)

 **Display:** the "Parking Lot Want to Know" instructions slide.

Do This: Revisit the Parking Lot from the warm up and facilitate a discussion around the "want to knows" students had.

Teaching Tip

If time allows, have students share responses from their graphic organizer. Find cause for celebration and praise in this mock exam. Encourage students to take pride in whatever they were able to accomplish, and to be ready to return in the next lesson ready to go!

Day Two (45 minutes)

Warm Up (5 minutes)

Teaching Tip

Revisit Parking Lot: You may want to spend time before the lesson or at the very beginning of the lesson reviewing the items added to the Parking Lot the day before, reminding students which themes were addressed the previous day and which will be addressed during the second day.

 **Display:** FRQ Annotation Strategies slide

Model: Use the next three slides to help students visualize the annotation and writing process they will use to answer the free response questions.

Teaching Tip

Do not spend overlong re-explaining these strategies. Students should have ample opportunity to put them to use during the mock-exam, wherein the teacher can monitor and assist as needed.

Activity (30 minutes)

 **Display:** The "You Should Have" slide.

Distribute: If collected last class, redistribute the PPR and activity guide.

Teaching Tip

Today you will answer the remaining two questions. Remember to use the annotation strategies we have been learning, and to pace yourself according to the priority you have set. You will have about 30 minutes to answer the remaining two questions.

Students who finish early should review their previous answers and revise as needed.

Circulate: Monitor students as they work, offering encouragement and reinforcing the use of the annotation strategies.

Wrap Up (10 minutes)

Reflection (5 min)

 **Display:** The reflections instructions slide.

Remarks

Flip to the last page of your activity guide and complete the three reflection questions. You can jot ideas, bullet points, or complete sentences.

Teaching Tip

Given the varied experiences students may have had during the exam, it's important to acknowledge that everyone's journey is unique. If some students face challenges, reassure them that struggling in these mock exams is a common part of the process. The aim of these practices is to gradually ease difficulties, fostering increased confidence for the actual exam day.

Glows, Grows, Want to Knows

 **Display:** The reflections instructions slide.

Discuss: *The areas students felt went well ("Glows"), areas to improve upon ("Grows"), and areas of confusion ("Want to Knows")*

Discussion Goal: Help guide students towards understanding the purpose of these practice exams, to give them an opportunity to prepare for the actual exam, and celebrate successes students had.

Teaching Tip

Extension Opportunity If time allows, the teacher can select one or more sample responses (either prepared ahead of time, or from actual student responses), identifies whether the response(s) is emerging, approaching expectations, or on target, and leads a whole-class discussion of the "glows" and "grows" for that response.



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